

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Navin Kumar - 19 (192511015) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin Kumar - 19

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1. Breadth - First search (BFS) and Depth - First search (DFS)

Breadth - First search (BFS)

- Breadth - First search (BFS) is an uninformed search algorithm that explores all nodes at the current depth before moving to the next level. It uses a Queue (FIFO)

Expansion Order: Arad → Zerind → Sibiu →

Timisoara → Oradea → Fagaras → Rimnicu Vilcea →

Lugoj → Bucharest

Solution path: Arad → Sibiu → Fagaras → Bucharest

Depth - First search (DFS)

Depth - First search (DFS) explores one branch completely before backtracking. It uses a

stack (LIFO)

Expansion Order: Arad $\rightarrow$ Zerind $\rightarrow$ Oradea $\rightarrow$ Sibiu

Fagaras $\rightarrow$ Bucharest.

Solution path: Zerind $\rightarrow$ Oradea $\rightarrow$ Sibiu $\rightarrow$ Fagaras

$\rightarrow$ Bucharest.

Comparison:

Feature	BFS	DFS
Data structure	Queue	stack
Search Method	level by level	Depth first
Completeness	Complete	Complete (finite graph)
optimality	yes	No
Memory usage	High	low
Time complexity	$O(b^d)$	$O(b^m)$
space complexity	$O(b^d)$	$O(b^m)$

Conclusion:- BFS is complete and optimal for

unweighted graphs because it always find

the shorter path. DFS uses less memory not

shorter path.

Tic-Tac-Toe playing agent

PEAS Description

Performance Measure

- Win the game
- Avoid losing
- Maximize the number of wins
- Force a draw if winning is impossible

Environment

- Place x or o in an empty square

Sensors

- Current board configuration
- Opponent's move
- Empty positions

Environment characteristics

- Fully observable: yes. Every board position is visible
- Deterministic: yes. Each move produces a predictable result

sequential: yes. Every move affect future move.

Not Episodic: Decisions depend on previous action.

static: yes, the board changes only when players make move

Discrete: yes, the board has a finite number of states and actions.

Multi-Agent: yes. Two players participate in game.

Why is it Adversarial?

The Tic-Tac-Toe environment is adversarial because two intelligent agents compete against each other. one player's success means the other player failure.

Conclusion:

The Tic-Tac-Toe agent operates in a fully observable, deterministic, sequential, static, discrete, and multi-agent environment. It is classified as adversarial because both players have conflicting goals.

A smart thermostat:-

Introduction:-

- A smart thermostat automatically maintain the desired room temperature. It can be designed as a simple Reflex Agent or a Model-Based Reflex Agent.

1. Simple Reflex Agent

The Agent makes decisions using only the current temperature.

Rules:-

- Current Temperature $<$ Target $\rightarrow$ Heater ON
- Current Temperature $>$ Target $\rightarrow$ Cooler ON
- Current Temperature $=$ Target $\rightarrow$ Heater & Cooler OFF

Limitations

- No memory
- cannot predict temperature changes
- Frequent ON/OFF switching

2. Model-Based Reflex Agent

The agent uses the current temperature along with an internal state to make better decisions.

Internal state maintained

- Current and previous temperature
- Target temperature
- Heater/cooler state
- Temperature trend

Performance Better

- Remembers previous readings.
- Predicts temperature changes
- Reduces unnecessary switching
- Save energy.

Conclusion:-

A smart thermostat automatically maintain and varied the room temperature.

And can be designed a simple

Reflex Agent or model based Reflex Agent.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand


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